Euclidean space = f.dim v sp/$\mathbb{R}$ with inner product $V \times V \to \mathbb{R}$

satisfying $\langle v, v \rangle > 0 \quad \forall v \neq 0$, $\langle v, w \rangle = \langle w, v \rangle$, $\langle cv_1 + v_2, w \rangle = \cdots$

$\langle v, cw_1 + w_2 \rangle = \cdots$

Main examples ① $\mathbb{R}^n$ with dot product

$$\left\langle \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} \right\rangle = \sum v_i w_i$$

② Any subspace of a Euclidean space

O.N. basis $e_1, \cdots, e_n$ means $\langle e_i, e_j \rangle = \delta_{ij}$

For any $v \in V$ $\quad v = \sum_i \langle e_i, v \rangle e_i$

Recall: Any set of mutually $\perp$ nonzero vectors is indep.

Def^n $\quad \|v\| = \sqrt{\langle v, v \rangle}$ "length/norm" $\quad$ "$v \perp w$": $\langle v, w \rangle = 0$ "perpendicularity"

Key tool For any subspace W, we have

$$V = W \oplus W^\perp \quad \text{—— (**)}$$

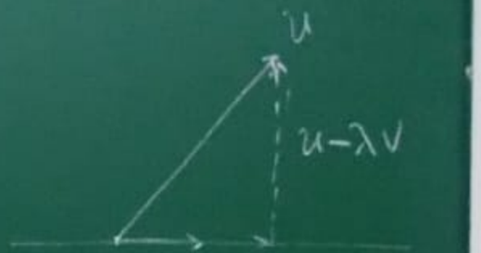
[For any subset S of V, $S^\perp := \{ v \mid v \perp s \ \forall s \in S \}$]

If $v \perp w$ then $\|v+w\|^2 = \|v\|^2 + \|w\|^2$ —— (*)

Formulas ① Orthogonal projection on a line $L = \mathbb{R}v$ (v fixed nonzero)

is a lin map $V \longrightarrow L$

$u \longmapsto \lambda v$ = "component of u along v"



How to find λ? $u = \lambda v + (u - \lambda v)$ "component of u that is $\perp$ to v"

$v \perp u - \lambda v$ gives $\langle v, u - \lambda v \rangle = 0 \Rightarrow \lambda = \dfrac{\langle v, u \rangle}{\|v\|^2}$

Pf 2 Let $v = \sum v_i e_i$, $u = \sum u_i e_i$ where $e_1, \dots, e_N$ O.N. basis. Bash it out: evaluate both sides and do algebra.

$$u = \frac{\langle v, u \rangle}{\|v\|^2} v + \left(u - \frac{\langle v, u \rangle}{\|v\|^2} v \right)$$

↑ Component along v ↑ Comp $\perp$ to v

② Cauchy Schwarz $|\langle v, u \rangle| \leq \|v\| \cdot \|u\|$ WLOG $v \neq 0$

Pf ⊛ $\Rightarrow \|u\|^2 \geq \left\| \dfrac{\langle v, u \rangle}{\|v\|^2} v \right\|^2 = \dfrac{|\langle v, u \rangle|^2}{\|v\|^4} \|v\|^2$ gives

$\text{LHS}^2 \leq \text{RHS}^2$, with equality iff $u \in \mathbb{R}v$

④ Gram-Schmidt process to get an o.N. basis $e_1, \cdots, e_n$ from given basis $v_1, \cdots, v_n$

$$e_1 = \frac{v_1}{\|v_1\|}$$

Observe: $\text{Span}\{v_1, v_2, \cdots v_k\} = \text{span}\{e_1, e_2, \cdots, e_k\}$ for $k = 1, \cdots, \dim V$.

$$e_2 = \frac{v_2 - \langle e_1, v_2\rangle e_1}{\|v_2 - \langle e_1, v_2\rangle e_1\|} \quad \text{check } \langle e_1, e_2\rangle = 0$$

$$e_3 = \frac{v_3 - \langle e_1, v_3\rangle e_1 - \langle e_2, v_3\rangle e_2}{\|v_3 - \langle e_1, v_3\rangle e_1 - \langle e_2, v_3\rangle e_2\|} \quad \langle e_1, e_3\rangle = 0, \ \langle e_2, e_3\rangle = 0$$

$$\vdots$$

$$e_{k+1} = \frac{v_{k+1} - \sum_{i=1}^{k} \langle e_i, v_{k+1}\rangle e_i}{\|\text{numerator}\|} \quad \underline{\text{Check}}\ \langle e_i, e_{k+1}\rangle = 0 \ \forall i = 1, \cdots, k$$

$v_k \in \text{span}\{e_1, \cdots, e_k\}$

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Thus change of basis matrix (either way) (P or P^{-1}) is upper Δ. Cor: If an operator T on an inner product space has an upper Δ matrix M, then it has an upper Δ matrix wrt an ON basis! (namely $P^{-1}MP$)

③ Δ inequality

$$\|v+u\| \leq \|u\| + \|v\|$$

Pf. $LHS^2 = \|v\|^2 + \|u\|^2 + \langle v,u\rangle + \langle u,v\rangle$

(by CS) $\leq \|v\|^2 + \|u\|^2 + 2\|u\|\|v\| = RHS^2$

Equality iff $\langle u,v\rangle = \|u\|\|v\| = \langle v,u\rangle$

which happens iff one of u,v is a nonnegative scalar multiple of the other

Ex $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$

Rmk Gram-Schmidt explicitly shows $V = W \oplus W^\perp$ (**)

Interesting operators on inner product spaces: ① "Orthogonal operator"

linear operator T preserving inner product

i.e. $\langle Tv, Tw\rangle = \langle v,w\rangle \quad \forall v,w \in V$

Rmks · In particular preserves perpendicularity

· T orthogonal $\Rightarrow \|Tv\| = \|v\|$ Ex Converse!

· So in particular injective & hence isom on f.dim. E.Sp

· Matrix of orthogonal T wrt an orthonormal basis $e_1, \dots, e_n$ has O.N. columns in $\mathbb{R}^n$ wrt dot product